

Recruiting Research Participants through Facebook: Assessing Facebook Advertisement Tools*

Anja Neundorf (University of Glasgow)

Aykut Öztürk (University of Glasgow)

July 26, 2021

Abstract

The use of paid advertisements on social media, in particular Facebook, to create samples for online survey research is becoming increasingly common. Facebook’s promise of unmediated, quick, and cheap access to a large pool of survey takers across the world is especially appealing to comparative political scientists and early career researchers. Yet the design of Facebook’s algorithm, geared towards business owners, complicates the use of Facebook for research purposes. In this paper, we rely on online surveys conducted in the United Kingdom, Turkey, Spain and the Czech Republic to reveal how two tools offered by Facebook, the choice of campaign objectives and the use of demographic targeting, affect the recruitment process, response quality, and sample characteristics. First, we demonstrate that the choice of campaign objectives can help researchers to distinguish *survey-takers* from *link-clickers* and *scrollers* on Facebook. Second, our analysis reveals how the use of targeting tools can produce balanced samples with small increases in the cost. Overall, we argue that balanced, inexpensive, and relatively high-quality samples can be created through Facebook only if researchers choose the right combination of Facebook tools, balancing efficiency gains with attention to the sample characteristics.

*We thank Thomas Gschwend, Francesco Rampazzo, and participants of 2021 PolMeth Europe Conference (virtual) for their constructive feedback.

Introduction: Potentials and Pitfalls

The use of social media as a tool for the recruitment of survey participants is becoming increasingly common among political scientists. Paid advertisements running on platforms such as Facebook and Instagram offer a unique opportunity for researchers, who need quick and cost-effective access to a pool of online survey participants. During the last few years, scholars have used Facebook Ads for recruitment in countries all around the world, such as Egypt (Williamson & Malik, 2020), Indonesia (Ananda & Bol, 2020), Kenya and Tanzania (L. R. Rosenzweig & Zhou, 2021), Thailand (Jäger, 2017), Turkey (Elci, 2021), and Uruguay (Bentancur et al., 2019). The Coronavirus pandemic has further revealed the potential of Facebook for scholarly research. Relying on Facebook Ads, researchers could quickly field surveys and continuously collect data from different countries, creating samples reaching tens of thousands respondents (Abu-Akel et al., 2021; Grow et al., 2020).

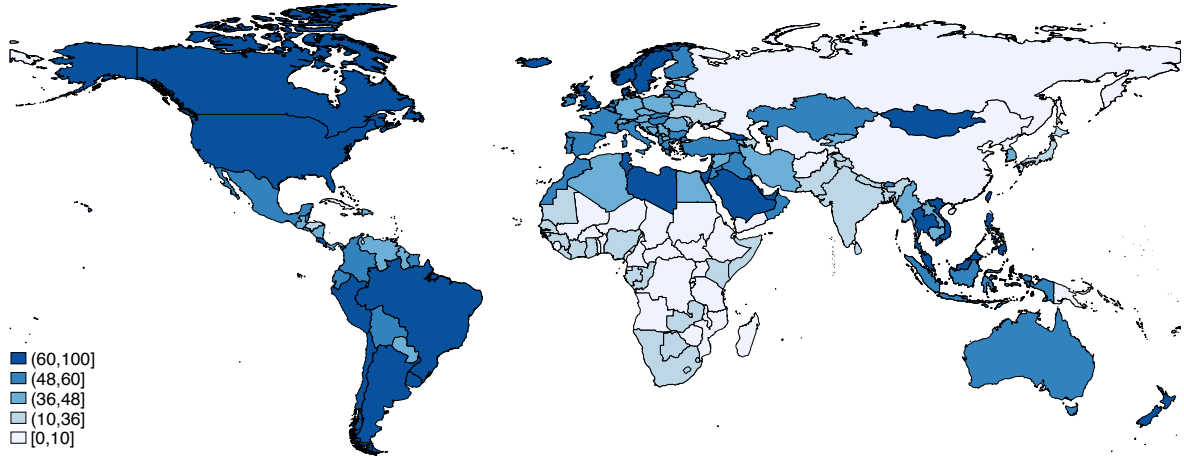
It is not surprising that it is easy to access large samples using Facebook, given the reach of the platform. Figure 1 presents the percentage of each country's population that is active on Facebook at least once per month. The data is from 2019 and totalled 2.5 billion, one third of the world population.¹ As the figure confirms in most countries - both democracies and autocracies, Facebook reaches a large proportion of the population, with notable exceptions in China, Russia, and many sub-Saharan countries.² Especially in Europe, North and South America, the Middle East and South East Asia, Facebook has extremely high penetration rates of over 50 percent of the population regularly using the platform. Given the easy access of researchers to an extremely large pool of potential study participants, we believe that Facebook is a highly valuable research tool.

While the use of paid advertisements on Facebook has a high potential to shape

¹Source: <https://worldpopulationreview.com/country-rankings/facebook-users-by-country>, accessed April 8, 2021.

²North Korea and Cuba are the only countries that have no active Facebook users.

Figure 1: Percentage of population that is active on Facebook at least once per month (2019)



data collection strategies during the next decade, scholars using Facebook need to pay special attention to the issues of sample biases, data quality and cost-effectiveness. First, like all other opt-in survey samples, Facebook samples suffer from the self-selection bias. Furthermore, since the platform is designed to serve business owners, rather than scholars, its algorithm usually directs advertisements to users who are most likely to click on them, aggravating the self-selection bias. Unless researchers directly intervene, most samples recruited through Facebook end up being significantly skewed towards a certain demographic group: sometimes towards men (e.g. Guiler, 2020) or women (e.g. Avenburg, 2019), sometimes towards older (e.g. Boas et al., 2020) or younger generations (e.g. Boas, 2014), and usually towards highly educated members of the society (L. Rosenzweig et al., 2020).

In addition to the issue of bias, cost-effectiveness of Facebook samples also requires attention. Low costs of Facebook samples is frequently listed as a reason to use Facebook Ads. It is important to remember that, however, cost-effectiveness is a function of Facebook tools selected to deliver advertisements and the advertisement content used to

draw Facebook users' interests. Facebook Ads utilize a system of competitive bidding among advertisers to assign limited advertisement spaces. While researchers can determine their overall advertisement budget and spending limits, they are never guaranteed a certain number of recruits from their advertisements for their budget. There are examples in which Facebook samples turn out to be significantly more expensive than other forms of recruitment (eg. Antoun et al., 2016; Gu et al., 2016).

In this paper, we systematically explore how the choice of two Facebook tools, campaign objectives and demographic targeting, affect the sample characteristics, response quality as well as the overall recruitment of participants, including costs. These tools are primary means through which scholars can influence Facebook's recruitment algorithm, and we compare eight different combinations of these tools to reveal how they affect the recruitment process. To the best of our knowledge, this is the first study to undertake an effort to compare the changes in sample composition, response quality, and recruitment costs occurring in response to researchers' choices of Facebook tools. To achieve these objectives, our paper relies on advertisement campaigns and surveys conducted in four countries, the United Kingdom, Turkey, Spain, and the Czech Republic, allowing us to draw more generalizable inferences.

We underline two key findings from this study. First, the use of conversion campaign objective allows researchers to directly reach Facebook users that will be more likely to complete an online surveys. This efficiency gain comes without any loss in sample balance or response quality. Second, Facebook produces very imbalanced samples if researchers do not use any targeting tools. Appropriate use of demographic targeting results in a significantly more balanced sample with a small increase in cost. Thus, the use of conversion campaign objective with extensive use of demographic targeting emerges as the best recruitment strategy for researchers using Facebook advertisements.

Facebook Tools: Customizing Facebook Algorithms for Research Purposes

Advertisers designing campaigns on Facebook are offered multiple tools to customize the advertisement campaign based on their needs and preferences. These Facebook tools determine which and how many Facebook users will see advertisements. Advertisers can choose the daily budget of their advertisements, through which devices advertisements will be delivered (mobile versus desktop) and on which platforms, such as Facebook, Instagram, and Messenger, advertisements are placed (Kühne & Zindel, 2020). In this study, we are keeping most of Facebook tools constant while focusing on two of them that we believe are especially relevant for concerns of academic research: campaign objectives and targeting.

Campaign objectives

Each researcher has to choose a campaign objective when using Facebook Ads Manager, which affects the menu of optimization strategies available to the advertiser. In this study, we compare the use of three campaign objectives, each representing a broader category of campaign objectives. According to Facebook's own descriptions, "awareness" refers to the goal of generating interest in products or services, "consideration" refers to the goal of getting people seek more information about the products, and "conversion" refers to the goal of encouraging people to buy or use advertisers' products. Facebook does not provide detailed information, however, on how its algorithm works to provide these results.

Researchers are not always explicit about which campaign objectives they are using, even when the article is about recruitment through Facebook (see, for example Kühne & Zindel, 2020). Based on our review of the literature, we discovered that researchers

have predominantly used “traffic” as their campaign objective, which is grouped under the “consideration” category of campaign objectives (e.g. Grow et al., 2020; Head et al., 2016; Jäger, 2017; Oesterle et al., 2018; Perrotta et al., 2021; Pötzschke & Braun, 2017; Samuels & Zucco, 2013; Schneider & Harknett, 2019; Zhang et al., 2020). This decision is sometimes justified with reference to the convention (Oesterle et al., 2018) but there is also a clear rationale to it. As Facebook describes it, “traffic” aims to send people from Facebook to any URL the advertiser chooses. In line with this goal, “traffic” offers “link clicks” as an optimization strategy. When “link clicks” is set as the optimization strategy, Facebook detects characteristics of individuals that are most likely to click on the business owner’s advertisements and then shows advertisements predominantly towards this group. This usually happens after a period of learning phase, which is around fifty link clicks (Zhang et al., 2020). Thus, “traffic” looks like a cost-effective strategy for many researchers whose goal is to bring more people to their survey page. Schneider and Harknett (2019, p. 7) report that they decided to use traffic, rather than reach or conversion campaigns, after consultation with advertising specialists at Facebook.

“Conversion,” grouped under the conversion category of campaign objectives, works on similar principles, but we see fewer examples of its use by researchers (see, Boas et al., 2020; Iannelli et al., 2020; L. Rosenzweig et al., 2020). This might partly be because it is newer and still unfamiliar to most researchers – it was introduced only in 2017; but this might also be because it is more complicated to use conversion objective – it requires generating a Facebook Pixel code unique to the advertiser and adding that code to the advertiser’s own website. With the help of this Facebook Pixel code, the conversion campaign objective lets Facebook track users’ behavior outside of the Facebook environment and optimizes accordingly. Thus, a researcher can use Facebook Pixel code to inform Facebook when a respondent successfully completes the survey and

reaches the “Thank You” page. After a learning phase, Facebook will then start showing advertisements to people who are more likely to complete the survey and not just click on the advertisement, as done using “traffic”.

One downside of both conversion and traffic strategies for research purposes is that optimization based on user behavior aggravates sample biases, leading to homogeneous samples. This problem has been frequently expressed in recent works (Grow et al., 2020; Munger et al., 2020; L. Rosenzweig et al., 2020; Zhang et al., 2020). Scholars pointing to these shortcomings aim to overcome these limitations using targeting. In this project, we also test if using the “reach” campaign objective, grouped under the “awareness” category, can alleviate this problem. Facebook promises that this campaign objective will maximize how many Facebook users will see an advertisement.

There are several reasons to test how Facebook campaign objectives work. First, most political scientists use “traffic” campaign objective, but we are not aware of any research that actually compares its use to other campaign objectives. Based solely on Facebook’s description, “conversion” emerges as a better fit for any advertisers who want Facebook users to complete some tasks or purchase a product. However, it is not always the case that Facebook tools deliver what they promise (see Sances, 2018), and we do not know if conversion algorithms will really work effectively when survey completion is defined as the task Facebook users need to complete. Furthermore, researchers need to also take into account balanced distribution of survey respondents across demographic groups and the easiness in setting up Facebook Ads Manager for scholarly research when deciding which Facebook campaign objective to use. A campaign objective that will recruit slightly fewer participants might still be preferable if it will produce significantly more balanced samples.

Targeting

Facebook allows researchers to target specific demographic groups with their advertisements. Targeting works especially well with age and gender since every Facebook user provides this information to Facebook themselves; however, it is less successful in meeting other targets, such as education and party, since these are usually derived from users' behaviors on Facebook (Sances, 2018).³ The use of targeted advertisements is becoming increasingly common among researchers using Facebook for survey recruitment (L. Rosenzweig et al., 2020; Zhang et al., 2020). In this study, our goal is to explore how two different ways of demographic targeting perform compared to no targeting and whether their effects interact with the campaign objectives listed above.

One reason to study demographic targeting is to document the trade-off between cost and representativeness across different ways of demographic targeting. This will allow researchers to make the best choice depending on their own research goals. In addition to this, we believe that it is also necessary to study targeting options in combination with campaign objectives. Demographic targeting requires creating more “advertisement sets,” each having their own budget. Each of these advertisement sets will focus on a smaller segment of population, by definition. For example, an advertisement set in Turkey not using any demographic targeting will target a population of 55,000,000 Facebook users while an advertisement set in Turkey targeting college educated older females will target only 1,000,000 Facebook users. Furthermore, the budget of the advertisement will now be smaller, as the overall campaign budget will be shared between advertisements separately targeting different demographic groups, such as non-college educated older females or college educated younger females. Importantly, each of these advertisement sets, targeting a smaller population with a smaller budget, will go through the learning process independently. This process directly affects Facebook’s conversion and

³Facebook points that “a disproportionate number of younger users register with an inaccurate age.” See: <https://sec.report/Document/0001326801-20-000013/fb-12312019x10k.htm> .

traffic optimization algorithms, using the feedback from traffic and conversion events to optimize. By exploring how different ways of demographic targeting interacts with campaign objectives, we can inform researchers better about trade-offs in selecting Facebook tools.

For this paper, we studied the use of two ways of demographic targeting in addition to no targeting. In our first online survey, conducted in the United Kingdom, we compared “none targeting” to “cross-targeting.” Under cross-targeting condition, we crossed subgroups of three age groups (young, middle-aged, old-aged), two gender categories (female, male) and three education categories (college graduates, high school graduates, and people with unspecified education status on Facebook). In total, we created 18 different advertisements, each having a £2 daily spending budget.⁴ Under “none targeting” condition, on the other hand, we assigned £36 to one single advertisement without defining any demographic target for that advertisement. Our concern here was to assign equal budgets to each of our campaigns.

In our subsequent studies in Turkey, Spain and Czech Republic, we added a new form of demographic targeting, which we call “single targeting.” In this way of targeting, we targeted each of eight groups listed above (young, middle-aged, old-aged, female, male, college graduates, high school graduates, people with unspecified education status on Facebook) separately. In total, we had eight different advertisements per campaign, each starting with a daily budget of £4.5. Compared to cross-targeting, advertisements with single targets have higher advertisement budgets and larger targeted populations. Thus, we thought, single-targeting could be a strategy bringing together the best of cross-targeting and non-targeting.

⁴£1 is the minimum daily advertisement budget. To make data collection faster, we decided to assign £2 per advertisement in cross-targeted campaigns.

Study Design

The ultimate goal of this study was to recruit participants to take part in an online survey experiment,⁵ which is a common research tool used by social scientists that utilize Facebook to recruit survey respondents (Ananda & Bol, 2020; Elci, 2021). More specifically, we investigate how Facebook Ads can be used to achieve this goal and how different Facebook tools to run these advertisements produce different sample characteristics, response quality and costs. The design of our study was pre-registered on the Open Science Foundation platform on February 8, 2021, before any data collection commenced.⁶

We conducted our study on Facebook tools in four countries: the United Kingdom, Turkey, Spain, and the Czech Republic. These four countries represent different cases, which are of interest for political scientists. The UK constitutes a typical case of an advanced democracy, which is the second most studied country according to Wilson and Knutsen (2020, p. 9). However, many researchers are more and more using Facebook ads to study less developed, non-democratic countries, where data collection is more restricted. Turkey hence represents such a context. Spain is located in the middle of these countries with respect to the level of political and economic development. Finally, Czech Republic is distinguished by its relatively small population, which amounts to only 10 million people. Small population should affect the recruitment process as it also means a smaller Facebook user base.

In our UK study, we included the comparison of six different Facebook advertisement campaigns, which were created through the combination of one type of campaign objective with the use of cross-group targeting or not using demographic targeting. Table 1

⁵The survey consisted of 26 questions and included a randomized survey experiments, dividing respondents into two treatment and one control group. The survey was about 10min long. Here we only present results based on the demographic questions as well as some political characteristics.

⁶The pre-analysis plan (PAP), the questionnaire, and the revision to the pre-analysis plan submitted after the data collection in the UK can be accessed here <https://osf.io/bfvsx/>.

Table 1: Modes of sample recruitment

COUNTRY: UNITED KINGDOM				
		Campaign Objectives		
		Conversion	Traffic	Reach
Targeting	None	<i>Sample 1</i>	<i>Sample 2</i>	<i>Sample 3</i>
	Cross	<i>Sample 4</i>	<i>Sample 5</i>	<i>Sample 6</i>
COUNTRIES: TURKEY, SPAIN, CZECH REPUBLIC				
		Campaign Objectives		
		Conversion	Traffic	
Targeting	None	<i>Sample 1</i>	<i>Sample 2</i>	
	Cross	<i>Sample 4</i>	<i>Sample 5</i>	
	Single	<i>Sample 7</i>	<i>Sample 8</i>	

lists these six modes of recruitment, or shortly campaigns, used in the United Kingdom.

We modified the study design after the UK survey. As we discuss in more detail below, campaigns using the “reach” campaign objective failed to recruit participants in the UK. After evaluating this to be a conclusive finding,⁷ we decided to use our research budget to explore how variations of targeting would affect the recruitment process. As such, we added two campaigns using “single-group targeting” to our Turkey, Spain, and Czech Republic studies to replace campaigns using the “reach” campaign objective. The bottom part of Table 1 presents the design of our studies in these three countries.

In terms of the sample size, our target was to continue recruitment for all campaigns until we reach at least 250 respondents in each sample to allow us to investigate the demographic characteristics of each sample. In the case of being unable to recruit 250 respondents in a particular sample, we decided to stop data collection for all campaigns once we reach 1,800 successful survey completions in total, which is 20% larger than the targeted sample size for this study. The goal of this stopping rule was to ensure that all campaigns in a given country would stay open for the same amount of time and spend

⁷We ran two pilot studies, one in the UK and one in Turkey, before the main data collection commenced. In both pilot studies we also noted that using the reach campaign failed to recruit survey respondents.

the same amount of budget.

In line with our expectations that some campaigns might not work well for survey participant recruitment, more than one campaign in all four countries failed to recruit the target 250 participants, forcing us to continue the data collection until we reached 1,800 successful participants per country in total. As planned, all advertisement campaigns from the same country stayed open for the same amount of time and spent the exactly same amount of budget, making advertisement campaigns comparable with other campaigns in the same country.

We used the same advertisement image and text for all advertisement campaigns, which was designed as a generic advertisement similar to the ones frequently used by previous researchers, which can be seen in Figure A.1 in the Appendix. No incentives were used for recruiting participants. Once participants saw our advertisements on their social media feed (on Facebook, Facebook Messenger, or Instagram), they decided whether to click the advertisement link or not. If they clicked on the advertisement, they were directed to the landing page of the survey, which was programmed in Qualtrics.⁸ They were then informed about the substantive survey and were asked to consent to their participation. Participants were then asked a series of demographic and political questions, which we analyze below. All participants successfully completing the survey were directed to a “Thank You” page hosted on a Wordpress web page. This webpage sent information to the Facebook Ads Manager about successful completions of “conversion” campaigns, allowing the Facebook algorithm to optimize towards completers. Our analysis below uses data collected through the Facebook Ads Manager (e.g. costs) as well as from the substantive survey that participants took.

⁸It was not possible for someone using the same device to take the survey multiple times, as Qualtrics allows researchers to use a so-called “prevent ballot box stuffing” function, which detects whether someone already completed a survey.

Findings

We summarize our findings below under two sections. First, we discuss how campaign objectives affect the recruitment process, response quality, and sample characteristics. Second, we discuss the use of demographic targeting and its effects on sample characteristics.

Campaign objectives: Distinguishing survey-takers from link-clickers and Facebook scrollers

The analysis in this section demonstrates that conversion is more cost-efficient than traffic and reach campaign objectives as it can help researchers to recruit Facebook users that are more likely to complete surveys without causing a significant loss in terms of representativeness and response quality of samples.

Recruitment process and campaign objectives

Table 2 presents results from conversion, traffic, and reach campaigns in all four countries. The analysis of campaign objectives reveals that Facebook delivers what it promises with respect to campaign objectives: Reach campaigns are shown to most Facebook users, traffic campaigns provide most link-clicks on the Facebook advertisement, and conversion campaigns produce the highest amount of successful survey completion.

To begin with, as can be seen in Row A in Table 2, our advertisements using the reach campaign objective were by far the ones seen by most Facebook users in the UK. Eleven times more people saw the ad when using “reach” compared to “conversion”. However, a very small percentage of these Facebook users, only 194 people, actually clicked on our links and only 6 of them completed the survey. We believe this is because Facebook algorithm shows advertisements under the reach category only to *Facebook scrollers*, i.e. Facebook users who do not usually click on links, keeping more engaged Facebook

Table 2: Comparison of campaign objectives

		Conversion	Traffic	Reach
United Kingdom (data collection: Feb 9-15, 2021)				
A	N of Facebook users who saw the ad on Facebook	56,486	82,480	655,815
B	N of Facebook users who clicked on the ad	1,597	1,598	194
C	N of respondents who completed the survey	1,032	756	6
b/a	<i>Link clickers/Facebook users seeing the ad (ctr)</i>	2.80%	1.90%	0.03%
c/a	<i>Completes/Facebook users seeing the ad</i>	1.80%	0.90%	0.00%
	Average costs per completed survey	£0.33	£0.44	£54.50
Turkey (data collection: Feb 19-22, 2021)				
A	N of Facebook users who saw the ad on Facebook	498,561	1,030,721	
B	N of Facebook users who clicked on the ad	4,603	17,906	
C	N of respondents who completed the survey	1,678	203	
b/a	<i>Link clickers/Facebook users seeing the ad (ctr)</i>	0.92%	1.74%	
c/a	<i>Completes/Facebook users seeing the ad</i>	0.34%	0.02%	
	Average costs per completed survey	£0.21	£1.76	
Spain (data collection: Mar 1-15, 2021)				
A	N of Facebook users who saw the ad on Facebook	314,494	985,910	
B	N of Facebook users who clicked on the ad	4,182	15,020	
C	N of respondents who completed the survey	1,709	92	
b/a	<i>Link clickers/Facebook users seeing the ad (ctr)</i>	1.32%	1.52%	
c/a	<i>Completes/Facebook users seeing the ad</i>	0.54%	0.01%	
	Average costs per completed survey	£0.87	£16.15	
Czech Republic (data collection: Mar 24 - Apr 7, 2021)				
A	N of Facebook users who saw the ad on Facebook	247,745	579,769	
B	N of Facebook users who clicked on the ad	3,949	12,068	
C	N of respondents who completed the survey	1,632	193	
b/a	<i>Link clickers/Facebook users seeing the ad (ctr)</i>	1.59%	2.08%	
c/a	<i>Completes/Facebook users seeing the ad</i>	0.66%	0.03%	
	Average costs per completed survey	£0.89	£7.51	

Note: Number of completed surveys per country: UK = 1,794; Turkey = 1,892; Spain = 1,801; Czech Republic = 1,827.

users for other campaign objectives. In our view the evidence is conclusive that “reach” does not work to recruit survey participants. We therefore did not use this campaign objective in other countries.

Traffic campaign objective, as advertised by Facebook, usually produced more link clicks on our advertisements compared to the other two campaign objectives. This can be seen by comparing the number of Facebook users who clicked on the ads in B rows of Table 2 and comparing the click-through rate (CTR) (B/A). While conversion and traffic

produced similar link clicks in the United Kingdom, in the remaining three countries traffic produced several times more link clicks with higher CTRs.

Importantly, however, successful survey takers formed only a small proportion of respondents who saw our advertisements through traffic campaign objectives, as can be seen in the C/A rows of Table 2. Facebook users recruited through traffic campaigns were more likely to leave our Qualtrics page immediately without answering the consent question; they were more likely to deny consent if they answered the consent question; and they were more likely to leave the survey incomplete if they gave consent.

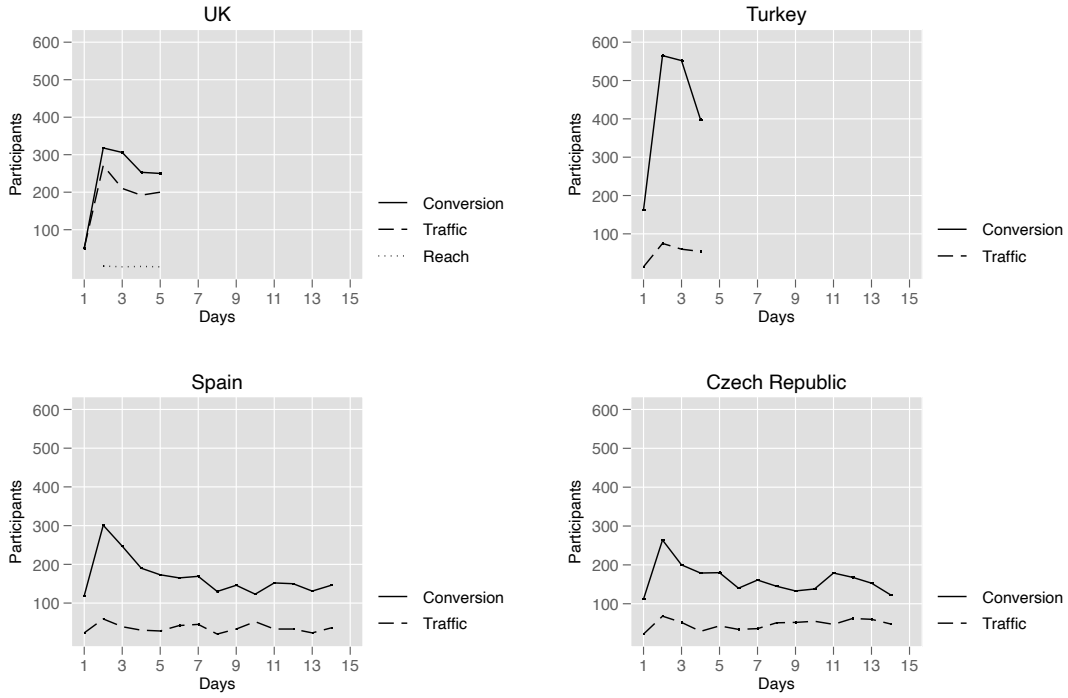
In addition to achieving more survey completions, we can further compare the average costs per respondent for each campaign objective. As Table 2 confirms, distinguishing survey-takers from link-clickers also provide cost-efficiency to our samples. For example, the average cost of a completed survey was 25% cheaper when conversion was used in the UK instead of traffic. The cost difference between the campaign objectives is even bigger in other countries, with traffic being 17 times more expensive than conversion in Spain for each completed survey.

The average costs were especially low in Turkey, where one completed survey costed around £0.21 when we used conversion as our campaign objective.⁹ In the UK, one completed survey can be as low as £0.25 (using non-targeted conversion), which is about 18 times lower than a recent similar online survey conducted by one of the authors in the US and UK with a commercial survey company, where the costs per respondent was £4.45. Even in Spain and the Czech Republic, where it costed significantly more to recruit participants compared to the UK and Turkey, a completed survey costed only £0.82 and £0.89 respectively when using conversion as the campaign objective.

The differences in average costs per country are driven by the number of days it took to reach the target sample size. Facebook will spend the daily budget as a default. The longer it is necessary to keep the ads open to recruit participants, the more expensive

⁹All prices reported here do not include sales taxes.

Figure 2: Number of participants recruited per day



Note: Data collection started at 5pm (GMT) on the first day of data collection; ads were closed at different times of the day during the last day as the sample size reached 1,800 participants. The graph does not report the number of recruited participants on the last day, as this varies by the time the survey was closed.

the overall data collection will be. Figure 2 plots the number of completed surveys per day for the different campaign objectives in our four cases.¹⁰ As the figures illustrate, the recruitment process was fastest in Turkey and the UK, which have larger Facebook user pools than Spain and especially the Czech Republic. This helped to reduce costs in the UK and Turkey. In all cases, conversion recruited more daily survey participants throughout the entire data collection.

Results regarding survey completion, costs and speed of participant recruitment from our four countries unequivocally demonstrate the superiority of conversion to recruit

¹⁰Figure A.2 in the Appendix further plots the number of completed surveys per day for each of the targeting tools, discussed in more detail in the next section.

survey participants. Although we spent the same amount of money for all campaigns, conversion campaigns always ended up with recruiting more successful survey completions. This was not because more Facebook users saw or clicked on the advertisements from conversion campaigns, but because participants recruited through conversion campaigns were always more determined to complete the survey.

The gap between the conversion and the traffic campaigns was clear even in our first survey in the UK. This difference further grew in the remaining countries to the extent that traffic campaign objective could produce very few successful survey completions at sometimes very high costs, despite providing a significant number of traffic to our Qualtrics page. We believe this is because Facebook’s tracking algorithm improved as we conducted surveys in different countries, successfully distinguishing “link-clickers” from Facebook users who would actually complete our surveys. In other words, Facebook used the budget we assigned for traffic campaigns to recruit link-clickers, while the budget for conversion campaigns was used to recruit survey-completers.

Sample characteristics and campaign objectives

Having demonstrated in the previous section, that recruitment of survey participants via Facebook is a cost-effective tool, we next ask whether cheaper prices via conversion are achieved at the cost of less-balanced sample characteristics. In other words, does conversion beat traffic and reach campaign objectives only because it is better at channelling resources to demographic groups that are better at completing tasks? In order to answer this question, Table 3 compares sample characteristics of traffic and conversion campaign objectives with population characteristics, taken from nationally representative surveys – the European Social Survey and Comparative Study of Electoral Systems.

Table 3 presents the proportion of demographic groups that we found harder-to-reach and who successfully completed the survey. We further conducted chi-square tests to compare the distributional properties of our samples with the population benchmark

Table 3: Comparing sample characteristics for conversion and traffic campaigns for survey completers

	Conversion	Traffic	Population Benchmark
United Kingdom			
Young aged (18-34)	7%***	4%***	29%
Middle aged (35-54)	17%***	11%***	34%
No college	39%***	44%***	70%
Female	42%***	44%***	51%
Less interested in politics	24%***	24%***	43%
Not partisan	42%***	47%	48%
Total	n=1,033	n=756	
Turkey			
Young aged (18-34)	21%***	19%***	37%
Middle aged (35-54)	38%	27%***	40%
No college	32%***	73%*	80%
Female	32%***	34%***	50%
Less interested in politics	33%***	54%	48%
Not partisan	42%***	56%***	32%
Total	n=1,689	n=203	
Spain			
Young aged (18-34)	10%***	5%***	22%
Middle aged (35-54)	29%***	20%***	41%
No college	43%***	57%***	73%
Female	45%***	36%**	52%
Less interested in politics	17%***	28%***	60%
Not partisan	22%***	34%	41%
Total	n=1,709	n=92	
Czech Republic			
Young aged (18-34)	35%***	20%	25%
Middle aged (35-54)	40%**	23%***	36%
No college	49%***	65%***	83%
Female	32%***	42%*	51%
Less interested in politics	24%***	51%***	82%
Not partisan	42%***	59%	61%
Total	n=1,634	n=193	

Note: Chi-square goodness of fit comparing Facebook and population samples: * = $p \leq 0.05$, ** = $p \leq 0.01$, *** = $p \leq 0.001$. The percentages presented here refer to the proportion of respondents in each of these demographics as a share of the overall sample in each campaign (column percentages). For example, 32% of respondents recruited through a conversion campaign in the UK are women (and conversely 68% are men). *Population benchmark:* We use data from the round 9 European Social Survey (for UK, Spain, and CZ) and from the wave 5 of Comparative Study of Electoral Systems (for Turkey) to create benchmark measures based on representative surveys.

samples. First of all, this table reveals some patterns that were common for all countries in our study. Our advertisement campaigns were less successful at recruiting younger, female, and non-college educated people. Furthermore, Facebook users completing our surveys had more interest in politics compared with the general population.

Importantly for our research question in this paper, demographic and political differences between conversion and traffic campaigns in our UK sample – the only sample in which traffic campaigns recruited a significant number of participants to allow for group comparisons and statistical analyses, seem to be quite small. Conversion samples fare even slightly better with respect to the age distribution, while traffic samples are slightly better at recruiting non-college graduates and females. Thus, preferring conversion to traffic campaign objective does not lead to a significant demographic or political bias.¹¹

Response quality and campaign objectives

Response quality is a common worry for samples recruited through social media. We use four metrics to measure response quality: passing an attention check, survey duration, engagement level with an open-ended question, and providing follow-up information at the end of the survey. These metrics provide a broad perspective on the response quality, measuring respondents' engagement level with the survey in addition to their attention levels. As many researchers are further interested to conduct follow-up surveys, the willingness of participants to share their contact details is important. The results of our quality checks are reported in Table 4, comparing these for two campaign objectives - conversion and traffic.

Attention checks are the most standard items that are used to check response quality.

¹¹Differences between conversion and traffic samples significantly grow in other countries as it can be seen on Table 3. Conversion is still better at recruiting younger people, but traffic samples are now less educated and less interested in politics, better reflecting population demographics. However, it is important to note that traffic samples in these countries are eight to eighteen times smaller than conversion samples, making the absolute number of less educated and female respondents still smaller than conversion samples. For example, conversion returns 523 non-college respondents in Turkey, while traffic returns only 119 respondents (at the same costs).

Table 4: Comparison of response quality for campaign objectives for survey completers

	UK		Turkey		Spain		Czech Rep.	
	Conv.	Traffic	Conv.	Traffic	Conv.	Traffic	Conv.	Traffic
Passed attention check	93%	92%	79%	52%	84%	83%	88%	72%
Median survey duration (in minutes)	6.6	7.4	8.5	6.4	7.5	9.6	6.4	7.3
Mean survey duration (in minutes)	12.6	9.5	11.7	7.8	10.3	11.8	12.8	16.1
Responded open-ended quest.	95%	95%	92%	65%	90%	89%	87%	85%
Words open ended quest.	17	18	12	7	15	15	12	10
Provided contact details	68%	61%	41%	22%	41%	27%	51%	39%

In this study, we used statements that asked participants to choose a particular response option. Respondents saw our attention check question only after completing 75% of the survey, and in each of our surveys this was the last statement in a three-items matrix question.¹² As such, our attention check measured whether respondents were still reading all of our questions as they approached the last parts of the survey. On average, 85% of our respondents passed this attention check, which is similar to the level reported in Alvarez et al. (2019)’s study using a commercial online panel and material incentives.¹³ As Table 4 demonstrate in all countries, respondents who came into the survey via conversion are always more likely to pass the attention check, compared to respondents who came into the survey via traffic. This is most pronounced in Turkey (27% difference) and the Czech Republic (16% difference).

Our second measure of response quality is survey duration. We generally expect that surveys completed in very short time periods mean low quality responses. Our initial expectation regarding survey duration, which was also mentioned in our advertisements, was ten minutes. Results were in line with our expectations: the mean value for survey duration was 11.5 minutes while the median value for survey duration was 7.3. As Table 4 shows, participants recruited through conversion spent on average more time to

¹²A screenshot of the attention check used in the UK is provided in Appendix A.3.

¹³Alvarez et al. (2019) use three attention checks in their survey. Their second and third attention checks are similar to ours, but the first one is different. Unfortunately, they drop respondents failing the first one so a direct comparison of proportions of inattentive responses in these two samples. However, based on their numbers, we can predict that the proportion of inattentives would be somewhere between 6% and 27%, if they started with the same attention check with ours.

complete the questionnaire.

Our third measure of quality refers to an open-ended question. In line with the substantial focus of our survey, we asked respondents to write down what they believe “democracy” means. Respondents had the option of skipping this question. In total, 90% of our respondents wrote at least one word, and the average response was 14 words long. The difference between conversion and traffic is less pronounced in terms of responding to the open-ended question in most countries, except in Turkey. Here respondents who came into the survey via conversion instead of traffic are significantly more likely to provide a response (92% versus 65%) and if they do, they provide more detailed answers (12 versus 7 words).

Finally, at the end of our survey we asked our respondents if they were willing to share their WhatsApp number or e-mail address, so that a re-contact would be possible, which is an indirect measure of response quality. We believe that a respondent will feel motivated to take another survey only if she is taking the current survey seriously. The proportion of participants providing follow-up information is also important for researchers who want to create panel data to measure long-term effects and changes. In total, 49% of our respondents provided their phone number or e-mail address. Given that not everyone uses WhatsApp or e-mail and people have privacy concerns, we believe that 49% shows a serious engagement with the survey¹⁴. In all countries, respondents who come into the survey via conversion are significantly more likely to provide contact details compared to traffic.

To summarize, conversion outperforms traffic and reach campaign objectives in terms of recruitment speed, cost efficiency, and in many of our response quality measures. We do not see major differences in terms demographic imbalances that could justify still using traffic as campaign objective. These findings lead us to the conclusion that scholars

¹⁴Kühne and Zindel (2020) report 62% success rate in collecting email addresses for their research on LGBTQ families in Germany

should use conversion campaign objectives when utilizing Facebook advertisements to recruit survey respondents.

Targeting tools

Our analysis in the previous section revealed that significant demographic imbalances might occur in both conversion and traffic samples: older, male, and more educated people are over-represented in our Facebook samples. Targeting tools provided by Facebook allow researchers to influence demographic and political characteristics of their sample. In this section, we discuss best practices to use these tools to provide more balanced and representative samples. Since we have already demonstrated a significant cost difference between conversion and traffic campaign objectives and since traffic campaign objectives failed to create meaningful samples in Turkey, Spain, and the Czech Republic, we limit our analysis in this section to samples recruited through conversion campaigns.

As mentioned above, we have defined three choices regarding targeting: No-targeting, single-group , and cross-groups targeting. In both the single-group and cross-groups targeting conditions, respondents were targeted based on age, sex, and education. In Table 5 we present the proportion of harder-to-reach groups within each of these demographic categories.¹⁵ We further conducted chi-square tests to compare the distributional properties of our samples with the population benchmark samples.

The first finding from this analysis relates to the trade-off between the sample size and the use of targeting tools. With the same amount of budget and time, less use of targeting tools usually returns larger samples. The size of the sample created through cross-targeting is only 52% of the sample created through none-targeting in the UK. Similarly, sizes of cross-targeting samples amount only to 70% to 80% of none-targeting sample sizes in other countries. Single-targeting, on the other hand, creates sample sizes

¹⁵We also present a figure comparing the performance of targeting conditions across countries in Appendix A.2.

Table 5: Single group distributions of participants recruited through conversion

	None	Targeting Single	Cross	Population Benchmark
United Kingdom				
Young aged	5%***	n/a	11%***	29%
Middle aged	11%***	n/a	28%**	34%
No college	34%***	n/a	48%***	70%
Female	39%***	n/a	49%	51%
Total	n=678	n/a	n=355	
Turkey				
Young aged	13%***	23%***	30%**	37%
Middle aged	41%	35%**	39%	40%
No college	20%***	33%***	45%***	80%
Female	25%***	33%***	38%***	50%
Total	n=613	N=597	N=479	
Spain				
Young aged	4%***	10%***	17%*	22%
Middle aged	23%***	26%***	40%	41%
No college	36%***	42%***	52%***	73%
Female	41%***	43%***	50%	52%
Total	n=584	n=658	n=467	
Czech Republic				
Young aged	38%***	36%***	29%*	25%
Middle aged	42%***	38%	38%	36%
No college	43%***	49%***	57%***	83%
Female	20%***	29%***	53%	51%
Total	n=629	n=553	n=452	

Note: Chi-square goodness of fit comparing Facebook and population samples: * = $p \leq 0.05$, ** = $p \leq 0.01$, *** = $p \leq 0.001$. The percentages presented here refer to the proportion of respondents in each of these demographics as a share of the overall sample in each of campaigns (column percentages). For example, 39% of respondents recruited through the non-targeted conversion campaign in the UK are women (and conversely 61% are men). *Population benchmark:* We use data from the round 9 European Social Survey (for UK, Spain, and CZ) and from the Comparative Study of Electoral Systems (for Turkey) to create benchmark measures based on representative surveys.

that are closer to none targeting, and even bigger in Spain.

A closer look at the Table 5, however, reveals why Facebook campaigns that do not use any targeting produce larger sample sizes: They are significantly more imbalanced than samples using cross-targeting. Although sample sizes are bigger in these campaigns, they usually end up recruiting fewer younger, female, and less educated respondents. This is because when demographic targeting is not used, Facebook algorithm is set free to

channel more resources to demographic groups that are more likely to complete surveys. This group is usually older, male and college graduate Facebook users.

The demographic imbalances in none-targeting campaigns become even clearer when we look at the intersection of three demographic properties – age, gender, and education, as presented in Table A.1 in Appendix. Facebook users that are older than 54, male, and college graduates form around 30% of none-targeting samples in the United Kingdom, Turkey, and Spain, although this demographic group corresponds to only 2% to 5% of the overall population in these countries. To the contrary, females that are younger than 55 and not holding a college degree form less than 1% to 5% of samples of none-targeting conditions despite forming 20% to 30% of the population in these countries. These imbalances are corrected to a great extent when scholars use targeting tools in general. As Table A.1 in the Appendix demonstrates, in samples formed through cross-targeting, all possible combinations of age, sex, and education are represented in the sample to a certain extent. The distribution of these groups within the sample are close to each other and to the distribution within the population, which we calculated based on representative high quality random samples.

A good convenient sample - as recruited using Facebook advertisement - should result in a large sample size with a balanced distribution across demographic groups.¹⁶ The latter is important not only because we need a certain number of people from each group to produce more generalizable inferences, but also because imbalances in demographic variables can correspond to imbalances in non-observable variables that are hard to control for in a regression analysis. Based on the analysis in this section, we conclude that systematic use of targeting tools is necessary to avoid samples greatly overrepresenting older-male-college graduate Facebook users, which might lead scholars to misleading results.

¹⁶It is worth however to keep in mind that social media users are not representative of the general population, as they are younger and better educated than non-users (Mellon & Prosser, 2017).

Conclusion

There is still some skepticism among political scientists over the use of online samples recruited through Facebook, rather than crowdsourcing platforms or online commercial panels (Krupnikov et al., 2021). Previous research has already demonstrated that Facebook can be used to produce high-quality samples with a fraction of the cost of commercial online panels (Zhang et al., 2020), but even more importantly, this skepticism misses the point that Facebook’s biggest promise is to comparative political scientists, whose research focus goes beyond the United States and other advanced countries. It is highly doubtful that commercial online panel companies or Amazon MTurk can provide a more diverse or higher quality sample in these countries than Facebook. In Turkey, for example, it took less than four full days for us to reach a sample of 1,800 participants, in which people from all age, education, and gender categories were represented. Benefiting from the worldwide reach of social media companies can help scholars overcome political behavior literature’s overwhelming focus on WEIRD nations - white, educated, industrialized, rich, and democratic.

The pitfall of recruiting through Facebook is that results are highly dependent on the choices made by scholars, and this is why political scientists need to conduct more studies exploring best practices to recruit through social media platforms. In this paper we explored two main tools used to set-up ads on Facebook - campaign objectives and demographic targeting. Our results show that researchers should not use the “reach” campaign, as this is not useful in participant recruitment. We further demonstrate that the most efficient campaign is “conversion”, which produced the largest, high quality and most cost-effective samples. We also documented how three different ways of demographic targeting affect the sample composition, concluding that the use of targeting is necessary to create more balanced samples although there is a small increase in the cost. Overall, the costs of using Facebook ads to recruit participants was extremely low,

especially compared to using commercial survey companies.

Given the potential of Facebook for comparative survey research, we believe that scholars should conduct more research exploring best practices for recruitment through Facebook. First, there are other Facebook tools that we kept constant and might affect the overall recruitment cost. For example, we kept our daily campaign budget constant at £36 per advertisement campaign, meaning that our total daily spending was always capped at £216 (as we had six separate samples in each country, as reported in Table 1). It is hard to tell how our results would change depending on different budget constraints. Bigger daily campaign budgets might have improved the learning process in conversion campaigns, further reducing costs. On the other hand, by publishing six campaigns at the same time in each of our countries, we created additional competition for the advertisement space, potentially increasing our overall costs. Smaller number of advertisement campaigns running at the same time would also decrease the costs.

Second, advertisement content was a constant in our study: We used the same advertisement image and text across all advertisement campaigns. We are not aware of any work by political scientists exploring how advertisement image and text affect the recruitment process or sample composition. Studies exploring these parts of Facebook advertisements are common in other disciplines, and especially in medical sciences (e.g. Bunge et al., 2019; Choi et al., 2017; Machado et al., 2019; Ryan et al., 2019; Sanchez et al., 2020). We believe political scientists should also explore how these choices affect our samples, especially with respect to political properties.

An important caveat to evaluate both our study and future studies on recruitment through social media is methodological challenges. One way to compare how Facebook advertisement campaigns fare against each other is to use Facebook’s A/B testing functionality, which lets advertisers change some variables of advertisements to determine which strategy performs best. We preferred not using it as we were not sure to what

extent Facebook would allow keeping variations of advertisements truly independent from each other. Instead, we manually created different advertisement campaigns with separate budgets and algorithms. Still, our conversion and traffic campaigns probably affected each other as they competed for similar audiences. Traffic campaigns could give better results if they did not have to compete against conversion campaigns because as conversion campaigns recognized “survey completers” and optimized towards them, traffic campaigns had to channel their resources towards “link clickers.” This might also be why reach campaigns failed so radically: As all “link clickers” and “survey completers” had been targeted by traffic and conversion campaigns, they were left with an unresponsive population of Facebook users. In addition to these, there might be some feedback effects between our country studies too. The progressive deterioration in the performance of traffic campaigns to recruit successful completes makes us think about this possibility. These concerns do not invalidate our findings, however; they rather demonstrate the success of conversion campaigns to perform well under the competition and their ability to improve their performances over time.

References

- Abu-Akel, A., Spitz, A., & West, R. (2021). The effect of spokesperson attribution on public health message sharing during the covid-19 pandemic. *PloS one*, 16(2), e0245100.
- Alvarez, R. M., Atkeson, L. R., Levin, I., & Li, Y. (2019). Paying attention to inattentive survey respondents. *Political Analysis*, 27(2), 145–162.
- Ananda, A., & Bol, D. (2020). Does knowing democracy affect answers to democratic support questions? a survey experiment in Indonesia. *International Journal of Public Opinion Research*.

- Antoun, C., Zhang, C., Conrad, F. G., & Schober, M. F. (2016). Comparisons of on-line recruitment strategies for convenience samples: Craigslist, Google AdWords, Facebook, and Amazon Mechanical Turk. *Field Methods*, 28(3), 231–246.
- Avenburg, A. (2019). Public costs versus private gain: Assessing the effect of different types of information about corruption incidents on electoral accountability. *Journal of Politics in Latin America*, 11(1), 71–108.
- Bentancur, V. P., Rodriguez, R. P., & Rosenblatt, F. (2019). Efficacy and the reproduction of political activism: Evidence from the broad front in Uruguay. *Comparative Political Studies*, 52(6), 838–867.
- Boas, T. C. (2014). Pastor Paulo vs. Doctor Carlos: Professional titles as voting heuristics in Brazil. *Journal of Politics in Latin America*, 6(2), 39–72.
- Boas, T. C., Christenson, D. P., & Glick, D. M. (2020). Recruiting large online samples in the United States and India: Facebook, Mechanical Turk, and Qualtrics. *Political Science Research and Methods*, 8(2), 232–250.
- Bunge, E. L., Taylor, L. A., Bond, M., Stephens, T. N., Nishimuta, K., Barrera, A. Z., Wickham, R., & Muñoz, R. F. (2019). Facebook for recruiting Spanish-and English-speaking smokers. *Internet interventions*, 17, 100238.
- Choi, I., Milne, D. N., Glozier, N., Peters, D., Harvey, S. B., & Calvo, R. A. (2017). Using different Facebook advertisements to recruit men for an online mental health study: Engagement and selection bias. *Internet Interventions*, 8, 27–34.
- Elci, E. (2021). Politics of nostalgia and populism: Evidence from Turkey. *British Journal of Political Science*.
- Grow, A., Perrotta, D., Del Fava, E., Cimentada, J., Rampazzo, F., Gil-Clavel, S., & Zagheni, E. (2020). Addressing public health emergencies via Facebook surveys: Advantages, challenges, and practical considerations. *Journal of Medical Internet Research*, 22(12), e20653.

- Gu, L. L., Skierkowski, D., Florin, P., Friend, K., & Ye, Y. (2016). Facebook, Twitter, & qr codes: An exploratory trial examining the feasibility of social media mechanisms for sample recruitment. *Computers in Human Behavior*, 60, 86–96.
- Guiler, K. G. (2020). From prison to parliament: Victimhood, identity, and electoral support. *Mediterranean Politics*, 1–30.
- Head, B. F., Dean, E., Flanigan, T., Swicegood, J., & Keating, M. D. (2016). Advertising for cognitive interviews: A comparison of Facebook, Craigslist, and snowball recruiting. *Social Science Computer Review*, 34(3), 360–377.
- Iannelli, L., Giglietto, F., Rossi, L., & Zurovac, E. (2020). Facebook digital traces for survey research: Assessing the efficiency and effectiveness of a Facebook ad-based procedure for recruiting online survey respondents in niche and difficult-to-reach populations. *Social Science Computer Review*, 38(4), 462–476.
- Jäger, K. (2017). The potential of online sampling for studying political activists around the world and across time. *Political Analysis*, 25(3), 329–343.
- Krupnikov, Y., Nam, H. H., & Style, H. (2021). Convenience samples in political science experiments. *Advances in Experimental Political Science*, 165.
- Kühne, S., & Zindel, Z. (2020). Using Facebook and Instagram to recruit web survey participants: A step-by-step guide and application. *Survey Methods: Insights from the Field (SMIF)*.
- Machado, N. M., Gomide, H. P., Bernardino, H. S., & Ronzani, T. M. (2019). Facebook recruitment of smokers: Comparing gain-and loss-framed ads for the purposes of an internet-based smoking cessation intervention. *Cadernos de saude publica*, 35, e00151318.
- Mellon, J., & Prosser, C. (2017). Twitter and Facebook are not representative of the general population: Political attitudes and demographics of British social media users. *Research & Politics*, 4(3), 2053168017720008.

- Munger, K., Luca, M., Nagler, J., & Tucker, J. (2020). The (null) effects of clickbait headlines on polarization, trust, and learning. *Public opinion quarterly*.
- Oesterle, S., Epstein, M., Haggerty, K. P., & Moreno, M. A. (2018). Using Facebook to recruit parents to participate in a family program to prevent teen drug use. *Prevention Science*, 19(4), 559–569.
- Perrotta, D., Grow, A., Rampazzo, F., Cimentada, J., Del Fava, E., Gil-Clavel, S., & Zagheni, E. (2021). Behaviours and attitudes in response to the COVID-19 pandemic: Insights from a cross-national Facebook survey. *EPJ data science*, 10(1), 1–13.
- Pötzschke, S., & Braun, M. (2017). Migrant sampling using Facebook advertisements: A case study of Polish migrants in four European countries. *Social Science Computer Review*, 35(5), 633–653.
- Rosenzweig, L., Bergquist, P., Pham, K. H., Rampazzo, F., & Mildenberger, M. (2020). Survey sampling in the Global South using Facebook advertisements.
- Rosenzweig, L. R., & Zhou, Y.-Y. (2021). Team and nation: Sports, nationalism, and attitudes toward refugees. *Comparative Political Studies*, 0010414021997498.
- Ryan, J., Lopian, L., Le, B., Edney, S., Van Kessel, G., Plotnikoff, R., Vandelanotte, C., Olds, T., & Maher, C. (2019). It’s not raining men: A mixed-methods study investigating methods of improving male recruitment to health behaviour research. *BMC Public Health*, 19(1), 1–9.
- Samuels, D. J., & Zucco, C. (2013). Using Facebook as a subject recruitment tool for survey-experimental research. *Available at SSRN 2101458*.
- Sances, M. W. (2018). Missing the target? using surveys to validate social media ad targeting. *Political Science Research and Methods*, 1–8.
- Sanchez, C., Grzenda, A., Varias, A., Widge, A. S., Carpenter, L. L., McDonald, W. M., Nemeroff, C. B., Kalin, N. H., Martin, G., Tohen, M., et al. (2020). Social me-

- dia recruitment for mental health research: A systematic review. *Comprehensive psychiatry*, 103, 152197.
- Schneider, D., & Harknett, K. (2019). What's to like? Facebook as a tool for survey data collection. *Sociological Methods & Research*, 0049124119882477.
- Williamson, S., & Malik, M. (2020). Contesting narratives of repression: Experimental evidence from Sisi's Egypt. *Journal of Peace Research*.
- Wilson, M. C., & Knutsen, C. H. (2020). Geographical coverage in political science research. *Perspectives on Politics*, 1–16.
- Zhang, B., Mildemberger, M., Howe, P. D., Marlon, J., Rosenthal, S. A., & Leiserowitz, A. (2020). Quota sampling using facebook advertisements. *Political Science Research and Methods*, 8(3), 558–564.

Online Appendix

Figure A.1: Facebook ad used in UK

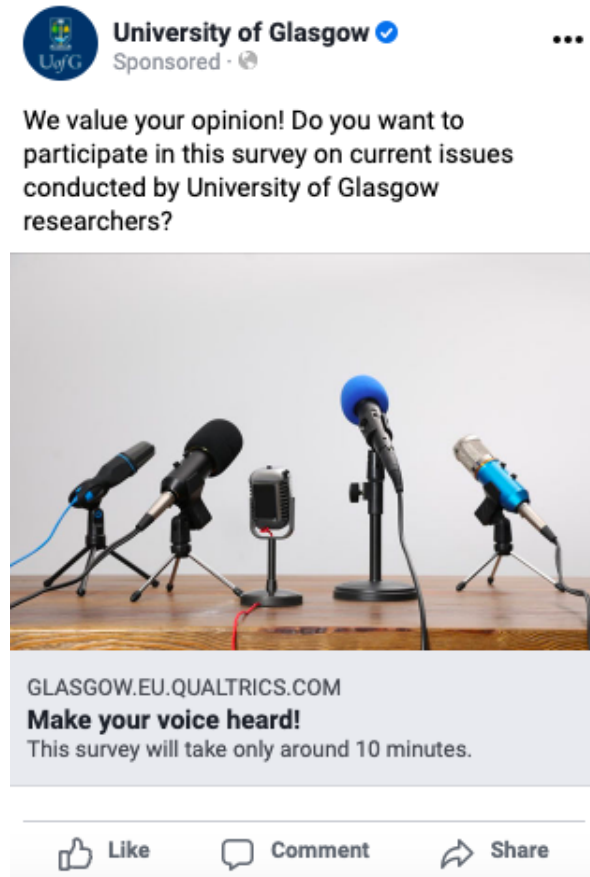


Figure A.2: Demographic targeting across countries and recruitment speed

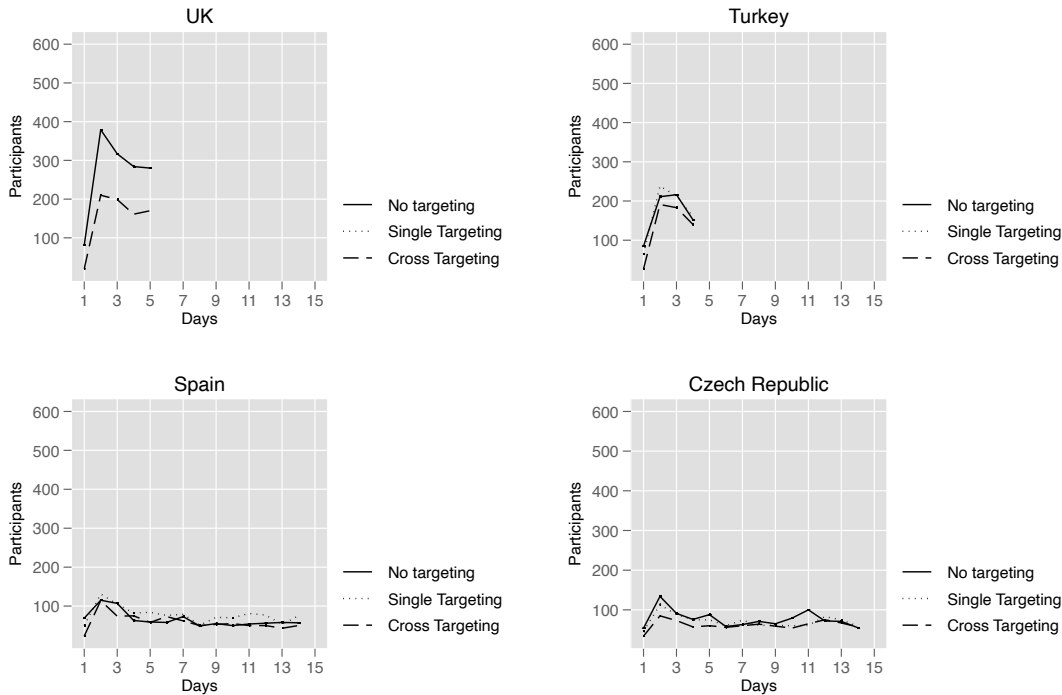


Figure A.3: Attention check

To what degree are you worried about the following situations?

	Not at all worried	Not much	A good deal	Very much worried	Don't know
Losing my job or not finding a job	☐	☐	☐	☐	☐
Paying for basics, such as food, bills, and rent	☐	☐	☐	☐	☐
For this statement, please choose "a good deal." This is for test purposes.	☐	☐	☐	☐	☐

Table A.1: Cross-group distributions of participants recruited through conversion by age, gender and education

Dem. group	UK			Turkey			Spain			Czech Republic		
	None	Cross		None	Single	Cross	None	Single	Cross	None	Single	Cross
Y, F, NG	5*** (1%)	5*** (1%)	8%	9*** (2%)	11*** (2%)	18*** (4%)	14%	4*** (1%)	6*** (1%)	16** (4%)	7%	13 *** (2%)
Y, M, NG	3*** (1%)	13*** (4%)	10%	6*** (1%)	31*** (5%)	43** (9%)	15%	5*** (1%)	10*** (2%)	19** (4%)	8%	21*** (4%)
Y, F, G	16** (2%)	5** (1%)	5%	24 (4%)	34 (6%)	27 (6%)	5%	4*** (1%)	14** (2%)	22 (5%)	4%	84 * (14%)
Y, M, G	11*** (2%)	16 (5%)	5%	38* (7%)	54*** (10%)	55*** (12%)	5%	6** (1%)	33** (5%)	22* (5%)	3%	67 (13%)
M, F, NG	4*** (1%)	18** (5%)	10%	10*** (2%)	14*** (2%)	28*** (6%)	17%	25*** (4%)	34*** (5%)	44 (10%)	12%	26 * (6%)
M, M, NG	13*** (2%)	20* (6%)	10%	41*** (7 %)	52*** (9%)	50** (11%)	17%	22*** (4%)	37*** (6%)	53 (12%)	13%	42*** (8%)
M, F, G	32 (5%)	35** (10%)	6%	53*** (9 %)	53*** (9%)	39*** (8%)	3%	48 (9%)	56 (9%)	47** (10%)	7%	27 *** (6%)
M, M, G	22** (3%)	25 (7%)	6%	139*** (24%)	81*** (14%)	58*** (13%)	4%	39 (7%)	43 (7%)	41** (9%)	6%	42*** (8%)
O, F, NG	62*** (9%)	53 (10%)	16%	6*** (1 %)	18*** (3%)	27 (6%)	8%	48*** (9%)	64*** (10%)	57** (12%)	18%	27 *** (6%)
O, M, NG	141*** (21%)	59* (17%)	13%	46 (8 %)	57 (10%)	40 (9%)	10%	104** (18%)	115** (18%)	47* (10%)	14%	32*** (7%)
O, F, G	146*** (22%)	55*** (16%)	4%	51*** (9%)	55*** (9%)	35*** (8%)	1%	109*** (19%)	105*** (16%)	45*** (10%)	3%	41 *** (10%)
O, M, G	216*** (32%)	47*** (13%)	5%	162*** (28%)	107*** (19%)	40*** (9%)	2%	161*** (28%)	125*** (20%)	46*** (10%)	4%	47 *** (11%)
Total (100%)	671	351		585	567	460		584	658	467		629
												553
												452

Note: Chi-square goodness of fit comparing Facebook and population samples: * = $p \leq 0.05$, ** = $p \leq 0.01$, *** = $p \leq 0.001$. Age = Young (Y), Middle-aged (M), Old (O). Gender = Female (F), Male (M). Education: Non-college (NC), College (C). Cells highlighted in bold refer to the campaign, which returned most participants in that particular demographic category in comparison to the other campaigns.

Setting-up a conversion campaign on Facebook Ads Manager - A user's guide

TBA!